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Robustness of Quantum Dot Power-Law Blinking

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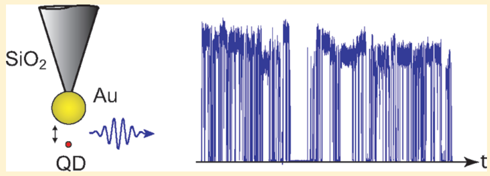
S Supporting Information

ABSTRACT: Photon emission from quantum dots (QDs) and other quantum emitters is characterized by abrupt jumps between an "on" and an "off" state. In contrast to ions and atoms however, the durations of bright and dark periods in colloidal QDs curiously defy a characteristic time scale and are best described by a power-law probability distribution, i.e., $\rho(\tau) \propto \tau^{-\alpha}$. We controllably couple a *single* colloidal QD to a *single* gold nanoparticle and find that power-law blinking is preserved unaltered even as the gold nanoparticle drastically modifies the excitonic decay rate of the QD. This resilience of the power law to change provides evidence that blinking statistics are not swayed by environment-induced variations in kinetics and provides clues toward the mechanism responsible for universal fluorescence intermittency.



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Colloidal semiconductor quantum dots (QDs) are promising candidates for a variety of applications in basic and applied science given their strong, stable and tunable fluorescence. They are, however, notorious for their blinking,[1] a term given to sudden jumps in fluorescence intensity from an "on" to an "off" state. The durations of dark periods (off-times) in colloidal QDs curiously defy a characteristic time scale and may range from microseconds to minutes for the same dot.[2] The off-times have a broad probability density distribution that follows a power law, i.e., $\rho(\tau_{\text{off}}) \propto \tau_{\text{off}}^{-\alpha_{\text{off}}}$. A similar distribution (with an exponential cutoff for large times) describes the on times as well.[2,3] The exponents α_{off} and α_{on} both range typically between 1 and 2.

While the exact mechanism responsible for QD blinking is still unclear, valuable clues can be inferred by identifying factors that affect or modify the power law behavior. In particular, the effect of the nanoenvironment on QD blinking has received wide attention,[4−9] but in some cases the findings are at variance with each other, e.g., Pelton et al. report that the power law does not change in different dielectric environments,[4] whereas others *do* see a systematic change.[10] Solution-based chemical approaches have yielded dramatic blinking suppressions, but not without differing opinions about the mechanisms involved.[6,8] Blinking suppression was also found for QDs near rough gold films[5,11,12] and on silver nanoprisms .[13] Interestingly, a modification of α_{off} was reported for QDs immobilized near silver islands.[7]

Most of the existing QD blinking studies make use of ensemble averaging or binning of *different* single QDs into the same histograms. These approaches are sensitive to artifacts due to variations in QD structure and local environments. For example, the plasmonic substrates used thus far to influence individual QDs have been geometrically ill-defined and have varied from laboratory to laboratory; sometimes the results have even shown significant variation from QD to QD on the same substrate within the same experiment.[5] These factors have precluded or obscured a systematic investigation of the mechanisms behind QD blinking (or its suppression). On a more fundamental note, QD blinking is known to exhibit weak ergodicity breaking.[14,15] This means that the ensemble average behavior deduced from observing n different QDs for T seconds each is *not* the same as the time average behavior inferred by looking at a single QD for nT seconds, not even in the limit of large n or T.

In this Letter, we examine for the first time the blinking statistics of a *single* QD interacting with a laser-irradiated gold nanoparticle at variable distance Δz (cf. inset of Figure 1). We emphasize that one and the same QD is used in a set of measurements, but the results have been verified for other QDs as well. A laser-irradiated gold nanoparticle functions as an optical antenna,[16] which concentrates incident radiation onto a single QD and helps the QD to emit radiation. An optical antenna in the form of a single spherical gold particle is a quantitative model system because of its simple and reproducible geometry.[17,18] The interaction strength of the nanoparticle and the QD can be tuned by continuously varying the separation between the two.

In our experiments, we attach a 80 nm gold nanoparticle to the end of a chemically functionalized glass tip.[19] The tip is attached to a tuning-fork crystal serving as a shear-force distance sensor for the antenna−QD separation.[18] QD samples are prepared by dispersing commercial colloidal CdSeTe/ZnS core shell QDs from solution (QDot705, Invitrogen) on a glass substrate. To identify individual QDs, we adjust the separation between the gold nanoparticle and the sample surface to ∼5 nm and raster scan the QD sample underneath the stationary nanoparticle while continuously detecting the fluorescence count rate and keeping the shear-force interaction constant via a feedback loop. QD blinking statistics are probed by stationing the gold nanoparticle

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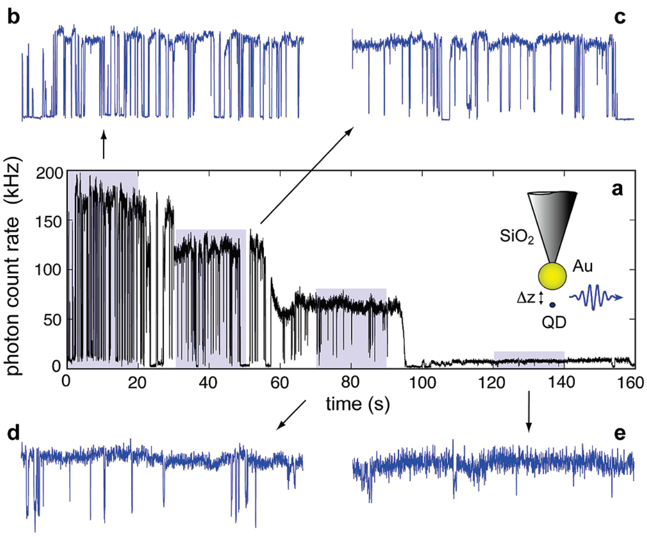

Figure 1. (a) Fluorescence time trace of a QD coming stepwise closer to a gold nanoparticle, starting $\Delta z \sim 7$ nm away and ending in "contact". Excitation intensity is ~200 W/cm^2 with a photon bin size of 10 ms. (b−e) Expanded views of equal time duration for the different separations. The QD blinking rate falls steadily with each step, dropping ~20× between (b) and (e). The inset illustrates the experimental configuration.

over a selected QD, adjusting the nanoparticle−QD separation to a fixed value, and time-binning the emitted fluorescence photons.

To avoid thresholding ambiguities in blinking analysis, we use QDs that show a largely binary switching behavior instead of a broad distribution of "on" (gray) states. The photon binning time is chosen to be large enough (typically between 1 and 50 ms) that the peak of the binned counts from the "on" state is separated from the background counts by more than 5 standard deviations. The binned photons are thresholded at two standard deviations above background to define "on" and "off" states. The distributions of "off" and "on" states are evaluated by binning their durations in unequal, but logarithmically evenly spaced, time windows. The number of occurrences (frequency) for each time bin are normalized with respect to the unequal bin sizes to yield probability densities. The resulting histograms are fitted with a single exponent power law using least-squares regression.

Figure 1 shows fluorescence time traces and blinking statistics for the *same* single QD but for different nanoparticle−QD separations, Δz. For a separation of Δz = 7 nm, we observe a fluorescence enhancement of a factor of ~10 relative to the fluorescence in absence of the gold nanoparticle. This enhancement is due to the local field enhancement, which leads to an increased excitation rate Γ_{exc} of the QD.[17,18,20] The fluorescence enhancement is accompanied by a reduction of the QD's excited state lifetime τ.[21] In the absence of the gold nanoparticle, we measure τ = 65 ns, and for Δz = 7 nm we obtain τ = 6.1 ns (see Supporting Information).

We next reduce the separation Δz in steps of ~2 nm and continue to record the fluorescence rate and the lifetime τ. As shown in Figure 1, the fluorescence rate drops at each step due to quenching (despite increased Γ_{exc}).[20] Moreover, the blinking rate appears to go down as Δz is reduced (Figure 1b−e). The QD stays predominantly "on" when the QD is in contact with the gold nanoparticle, with very few blinks (Figure 1e). This suggests a blinking suppression mechanism involving strongly modified decay rates (cf. ref 5).

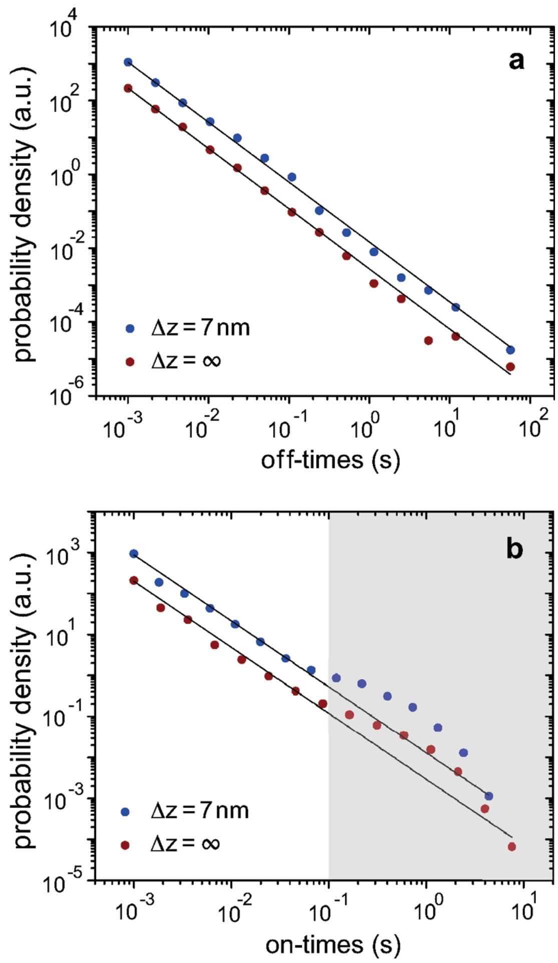

Figure 2. QD blinking statistics evaluated for different separations Δz between QD and gold nanoparticle. (a) Off-time distributions with and without coupling to the gold nanoparticle. There is no change in the slope ($\alpha \approx 1.62$). (b) Corresponding on-time distributions. Again, no change in the slope (≈1.60) is observed. The external laser intensity was reduced by a factor 10 when the antenna was brought close to the QD to compensate for the increase in local excitation rate Γ_{exc}.

The fluorescence time traces shown in Figure 1 indicate that long fluorescence off-times are less likely for small Δz and that the blinking statistics becomes modified. However, our measurements reveal that there is no change in blinking statistics, even for QDs in direct contact with a gold nanoparticle. Figure 2a shows off-time histograms for the same QD evaluated in presence and in absence of the gold nanoparticle. The histograms were evaluated from 100 s long fluorescence time traces and are characterized by the well-known power-law behavior. Interestingly, we observe no difference in the exponent α_{off}; that is, the off-time distribution of the QD remains unaffected by the proximity of the gold nanoparticle. For the particular QD used in Figure 1, α_{off} remained constant at around 1.62 (Figure 2a). Similar off-time behavior was seen for other QDs as well: in every case the change in the exponent was less than 10%. The on-time histograms show a characteristic shoulder around 0.1 s (seen across all QDs studied) followed by the exponential cutoff,[22] but the power law exponent for short times α_{on} again remains unaffected by the gold nanoparticle (Figure 2b). We have evaluated on and off time distributions for many different separations Δz between QD and

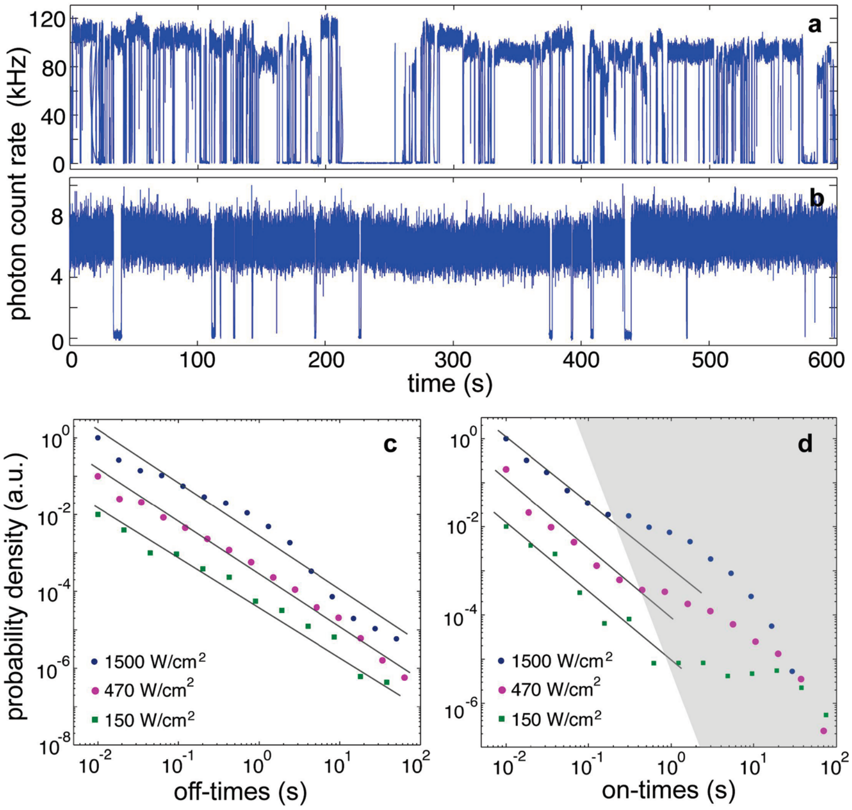

Figure 3. Blinking statistics as a function of excitation intensity. Fluorescence time traces acquired with an intensity of 1500 W/cm^2 (a) and 150 W/cm^2 (b) show markedly different blinking behavior, indicating that the blinking probability is proportional to the excitation rate Γ_{exc}. The photon binning window was fixed at 10 ms. (c) Off-time distributions for three excitation intensities. The exponents yield α_{off} (1.40 ± 0.15). (d) Corresponding on-time distributions also show no change in the slope ($\alpha_{on} \approx 1.55$), but as shown by the shaded gray box, the onset of the characteristic shoulder (and cutoff) steadily shifts to higher on-times as the excitation rate is reduced. The distributions in (c) and (d) are offset vertically for clarity.

gold nanoparticle, even for $\Delta z \approx 0$ for which the fluorescence emission is quenched by 97%, and in all cases we find no significant change in the power-law exponents. Our observations appear to be in contrast to previous studies that reported a significant change in the off-time exponent α_{off} near metal nanostructures.[5,7] We believe that the observed change in α_{off} is due to the fact that different QDs have been compared with each other.

A visual inspection of the time traces displayed in Figure 1 suggests that the blinking rate is reduced for small Δz. This observation appears to be contradictory, at first sight, to our result that there is no appreciable change in the power-law exponents α_{off} and α_{on}. Note, however, that a blinking rate in terms of blinks per unit time is strictly speaking ill-defined for a nonergodic process like QD blinking, since it depends on the observation time. The average from a finite time trace is therefore not representative for the long time behavior, which is significantly influenced by very long on or off durations. Second, the apparent reduction of the blinking rate for small Δz is at the expense of a reduced fluorescence intensity and hence the time that the QD spends in the excited state.

To understand the influence of the fluorescence intensity on the blinking statistics, we once again identified an isolated QD and recorded fluorescence time traces for three different excitation powers: 1500, 470, and 150 W/cm^2. Panels a and b of Figure 3 depict 600 s long time traces for the two extreme intensities, showing that the frequency of blinks goes down as the fluorescence rate is reduced. To determine the corresponding statistics of fluorescence intermittency for varying excitation intensities, time traces of different durations (600, 1200, and 1800 s) were recorded to yield a comparable number of blink events for each case. As shown in Figure 3c, the distribution of off times remains largely unchanged. The exponential cutoff in on times is shifted to longer times as the excitation intensity goes down, but the slope α_{on} does not change. These observations agree with the findings of Shimizu et al. who investigated the excitation power and temperature dependence of QD blinking.[22] The main conclusion of these authors was that lower temperatures and lower excitation intensities both lead to fewer blinking events but with no change in α_{off} or α_{on}. The longer average on times are, however, reflected as a shift in the exponential cutoff in the on-time distribution toward longer times.

We have also investigated the influence of local electrostatic potentials on the blinking statistics. A tungsten tip (<100 nm in diameter), fabricated by electrochemical etching, was positioned over a selected QD deposited on a conducting indium−tin oxide (ITO) coated glass substrate, and a potential has been applied between the tip and the ITO. The selected QD has then been exposed to high static electric fields (on the order of 10^8 V/m). The resulting distributions for off- and on-times remained largely unaffected by the local potential (see Supporting Information), indicating that strong static electric fields have at best only a small effect on the distribution of off times.[7]

Our results can be explained within the framework of a diffusion-controlled electron transfer (DCET) model[22−24] illustrated in

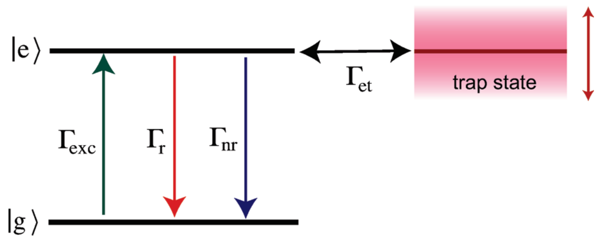

Figure 4. "Competing rates" model for QD blinking near a gold nanoparticle. The nanoparticle has a strong influence on the radiative and nonradiative decay rates (Γ_r and Γ_{nr}) while the intrinsic energy transfer rate (Γ_{et}) and spectral diffusion of the trap remain largely unchanged.

Figure 4. The model assumes that a bright QD goes dark when a photoexcited carrier is trapped.[25−27] Moreover, the trap energy itself diffuses in time, and transitions to and from the trap occur (at a rate Γ_{et}) only when the trap energy is aligned with an electronic level. Power-law distributed "on" and "off" durations (with exponents $\alpha_{off/on}$ around 1.5) then follow naturally in this model from the distribution of times between successive crossings of the origin in a random walk, also known as first passage times. For an "on" → "off" transition to occur, the levels need to be aligned and the QD needs to be in the photoexcited state. When these conditions are met, there are three possibilities for the QD to transition away from the excited state, namely, by radiative or nonradiative decay to the ground state (rates Γ_r and Γ_{nr}) or by energy transfer to the trap state (rate Γ_{et}). Only the latter gives rise to fluorescence blinking. Hence we can represent the blinking probability Q_{bl} as

$$Q_{bl} = \Gamma_{et}/[\Gamma_{et} + \Gamma_r + \Gamma_{nr}] \tag{1}$$

In the perturbation limit, the blinking rate Γ_{bl} (number of blinks per unit time) is simply the product of Q_{bl} and the cycling or excitation rate Γ_{exc}

$$\Gamma_{bl} = Q_{bl}\Gamma_{exc} = \frac{\Gamma_{exc}\Gamma_{et}}{\Gamma_{et} + \Gamma_r + \Gamma_{nr}} \tag{2}$$

While the gold nanoparticle increases Γ_{exc}, Γ_r, and Γ_{nr}, we argue that it leaves Γ_{et} largely unaffected. Furthermore, because of reciprocity, the enhancements of Γ_{exc} and Γ_r are similar,[16] and hence it follows that the reduction of Γ_{bl} for very short QD−nanoparticle separations is a result of the strong increase of Γ_{nr}/Γ_r,[17] and thus a drop of the fluorescence quantum yield. Meanwhile, blinking *statistics* continues to be governed by diffusive dynamics in the background and remains unaffected by the nanoparticle. We note that the DCET model can accommodate a deviation of the exponent from 1.5 by introducing a time-dependent diffusion constant.[23]

Let us stress the difference between our suggested mechanism and what has been proposed before,[5−7] wherein an increased decay rate boosts the quantum yield in the dark state up to a value comparable to that in the bright state. The mechanism we propose effectively slows down the "on" → "off" transition rather than influencing exciton dynamics in the "off" state. Our work is reminiscent of early work by Kuhn on reduced photobleaching of chromophores near metal surfaces due to quenching.[28] Another manifestation of the same principle appears in a recent study that reported modified blinking of single molecules close to a metal film.[29]

In conclusion, we coupled an optical antenna in the form of a spherical gold nanoparticle to an isolated QD, and recorded a 10-fold increase in fluorescence due to local field enhancement. The blinking rate was found to be drastically suppressed at the expense of a much higher nonradiative exciton decay rate (e.g., QD in direct contact with nanoparticle), but the influence on blinking statistics ("on" and "off" time distributions) was remarkably negligible. Our results suggest that the gold nanoparticle affects blinking kinetics by increasing the decay rates, but does not influence the physical parameters involved in blinking (e.g., trap energies or their diffusion).

■ ASSOCIATED CONTENT

Ⓢ Supporting Information. Additional information regarding lifetime changes due to coupling to an optical antenna, influence of local static fields, and fluorescene time traces and statistics for short antenna−QD distances. This material is available free of charge via the Internet at http://pubs.acs.org.

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